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Develop a Progressive Web Application (PWA) that supports offline functionality using a Service Worker and the cache-first caching strategy.

Project type: Progressive Web Application

Steps to perform

1. Create an ASP.NET Core MVC project or a suitable static web project in Visual Studio.

2. Add HTML or Razor pages, manifest.json, service-worker.js, and any referenced icons under wwwroot.

3. Register the service worker and link the manifest in the HTML layout.

4. Run the site on localhost or HTTPS and inspect it in a modern browser.

5. Use browser Developer Tools to check manifest, service worker, cache and offline operation. Push notifications require additional permission and subscription/server setup.

Closest supplied practical: </body> </html> Develop a PWA that works in offline mode using Service Worker

Source: Dot_Net_Practicals_6_to_10_Organized_v2.pdf

Provided creation instructions and source files

```
Creation steps: Create an MVC project named ServiceWorkerPWA. Add the home view, manifest and service
worker; link the manifest and register the service worker in the shared layout.
Filename: Index.cshtml    Location: Views/Home/Index.cshtml
<div class="text-center">
    <h1>College Information Portal</h1>
    <h3>PWA Offline Demo</h3>
    <p>This application works without internet connection.</p>
</div>
Filename: manifest.json    Location: wwwroot/manifest.json
{
  "name": "Offline PWA",
  "short_name": "OfflinePWA",
  "start_url": "/",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "192x192",
      "type": "image/png"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
Filename: service-worker.js    Location: wwwroot/service-worker.js
const CACHE_NAME = "pwa-cache-v1";
const urlsToCache = [
    "/",
    "/Home/Index"
];
self.addEventListener("install", event => {
    event.waitUntil(
        caches.open(CACHE_NAME)
            .then(cache => {
                return cache.addAll(urlsToCache);
            })
    );
});
self.addEventListener("fetch", event => {
    event.respondWith(
        caches.match(event.request)
            .then(response => {
                return response || fetch(event.request);
            })
    );
});
Filename: Layout    Location: Views/Shared/_Layout.cshtml
<!DOCTYPE html>
```

```
<html lang="en">
<head>
    <meta charset="utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>@ViewData["Title"] - _SWOfflinePWA_</title>
    <script type="importmap"></script>
    <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
    <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
    <link rel="stylesheet" href="~/_SWOfflinePWA_.styles.css" asp-append-version="true" />
    <link rel="manifest" href="/manifest.json">
</head>
<body>
    <header>
        <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-white border-bottom
box-shadow mb-3">
            <div class="container-fluid">
                <a class="navbar-brand" asp-area="" asp-controller="Home"
asp-action="Index">_SWOfflinePWA_</a>
                <button class="navbar-toggler" type="button" data-bs-toggle="collapse" d

ata-bs-target=".navbar-collapse" aria-
controls="navbarSupportedContent" aria-expanded="false" aria-label="Toggle navigation">
                    <span class="navbar-toggler-icon"></span>
                </button>
                <div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">
                    <ul class="navbar-nav flex-grow-1">
                        <li class="nav-item">

                            <a class="nav-link text-dark" asp-area="" asp-controller="Home"
asp-action="Index">Home</a>
                        </li>
                        <li class="nav-item">
                            <a class="nav-link text-dark" asp-area="" asp-controller="Home"
asp-action="Privacy">Privacy</a>
                        </li>
                    </ul>
                </div>
            </div>
        </nav>
    </header>
    <div class="container">
        <main role="main" class="pb-3">
            @RenderBody()
        </main>
    </div>
    <footer class="border-top footer text-muted">
        <div class="container">
            &copy; 2026 - _SWOfflinePWA_ - <a asp-area="" asp-controller="Home"
asp-action="Privacy">Privacy</a>
        </div>
    </footer>
    <script src="~/lib/jquery/dist/jquery.min.js"></script>
    <script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
    <script src="~/js/site.js" asp-append-version="true"></script>
    @await RenderSectionAsync("Scripts", required: false)
    <script>
        if ('serviceWorker' in navigator) {
            navigator.serviceWorker.register('/service-worker.js')
                .then(() => console.log('Service Worker Registered'));
        }
    </script>
</body>
</html>
```

CHECK: This is a related source example, not necessarily a complete or exact solution to the wording above. Adapt and verify it.

Expected result / verification

The application should meet the exact question requirements. Record actual output or screenshots after running; no execution has been claimed in this manual.